

Customer Shopping Behavior Analysis

1. Project Overview

This project analyzes customer shopping behavior using transactional data. The aim is to uncover insights into spending patterns, customer segments, product preferences, and discount usage to support business decision-making.

2. Dataset Summary

- **Total Rows:** 3,900
- **Total Columns:** 18

Key Features:

- Customer segment : Age, Gender, Location, Subscription Status
- Purchase details : Item Purchased, Category, Purchase Amount, Discount Applied
- Shopping behavior : Previous Purchases, Frequency, Review Rating, Shipping Type.
- Missing Data : 37 values in Review Rating Column.

3. Exploratory Data Analysis Using Python

Data Preparation Steps

- First imported the dataset using **pandas**.
- The dataset was successfully loaded and inspected for structure and summary statistics using **df.info()** and **df.describe()**.
- Missing values in the Review Rating column were identified.

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	3900	3900
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	2	2
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	No	No
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	2223	2223
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN	NaN	NaN
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN	NaN	NaN
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	NaN	NaN
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	NaN	NaN
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	NaN	NaN
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	NaN	NaN
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	NaN	NaN

Previous Purchases	Payment Method	Frequency of Purchases
3900.000000	3900	3900
NaN	6	7
NaN	PayPal	Every 3 Months
NaN	677	584
25.351538	NaN	NaN
14.447125	NaN	NaN
1.000000	NaN	NaN
13.000000	NaN	NaN
25.000000	NaN	NaN
38.000000	NaN	NaN
50.000000	NaN	NaN

Missing Data Handling

- Missing values in the **Review Rating** column were handled by imputing the **median rating of each product category**, ensuring data consistency while avoiding bias from extreme values.

Column Standardization

- Column names were standardized to **snake_case** format to improve readability and maintain consistency.

Feature Engineering

Additional features were created to support deeper analysis:

- Created **age_group** column customers were grouped into age brackets.
- Created **purchase_frequency_days** column from purchase frequency information.

Data Consistency Check

- A comparison between **discount_applied** and **promo_code_used** confirmed redundancy.
- As both columns represented the same information, **promo_code_used** was removed to avoid duplication.

Database Integration

The cleaned dataset was successfully loaded into a relational database for structured SQL analysis.

4. Data Analysis Using SQL

SQL was used to answer key business questions based on transactional data.

Revenue by Gender: This analysis compares total revenue generated by male and female customers.

Observation:

- Male customers generated significantly higher total revenue compared to female customers.

	gender	Total_Rvenue
▶	Male	157890
	Female	75191

Top 5 Products by Rating : This analysis highlights products with the highest average customer review ratings.

Observation:

- Products such as **Sandals, Boots** and **Handbags** received consistently higher ratings, indicating strong customer satisfaction.

	item_purchased	Highest_Review_Rating
▶	Sandals	3.82
	Boots	3.79
	Handbag	3.78
	Hat	3.78
	Gloves	3.78

Shipping Type Comparison : Average purchase amounts were compared between Standard and Express shipping.

Observation:

- Customers opting for **Express shipping** tend to spend slightly more than those choosing Standard shipping.

	shipping_type	Average_Purchase_Amount
▶	Express	60.48
	Standard	58.46

Subscribers vs Non-Subscribers : This analysis compares Total customers, Average spend and Total revenue across subscription status.

Observation:

- **Non-subscribers** contribute higher total revenue due to larger customer volume, while average spend remains similar across both groups.

	subscription_status	Total_Customers	Total_Revenue	Average_Spend
►	No	2847	170436	59.87
	Yes	1053	62645	59.49

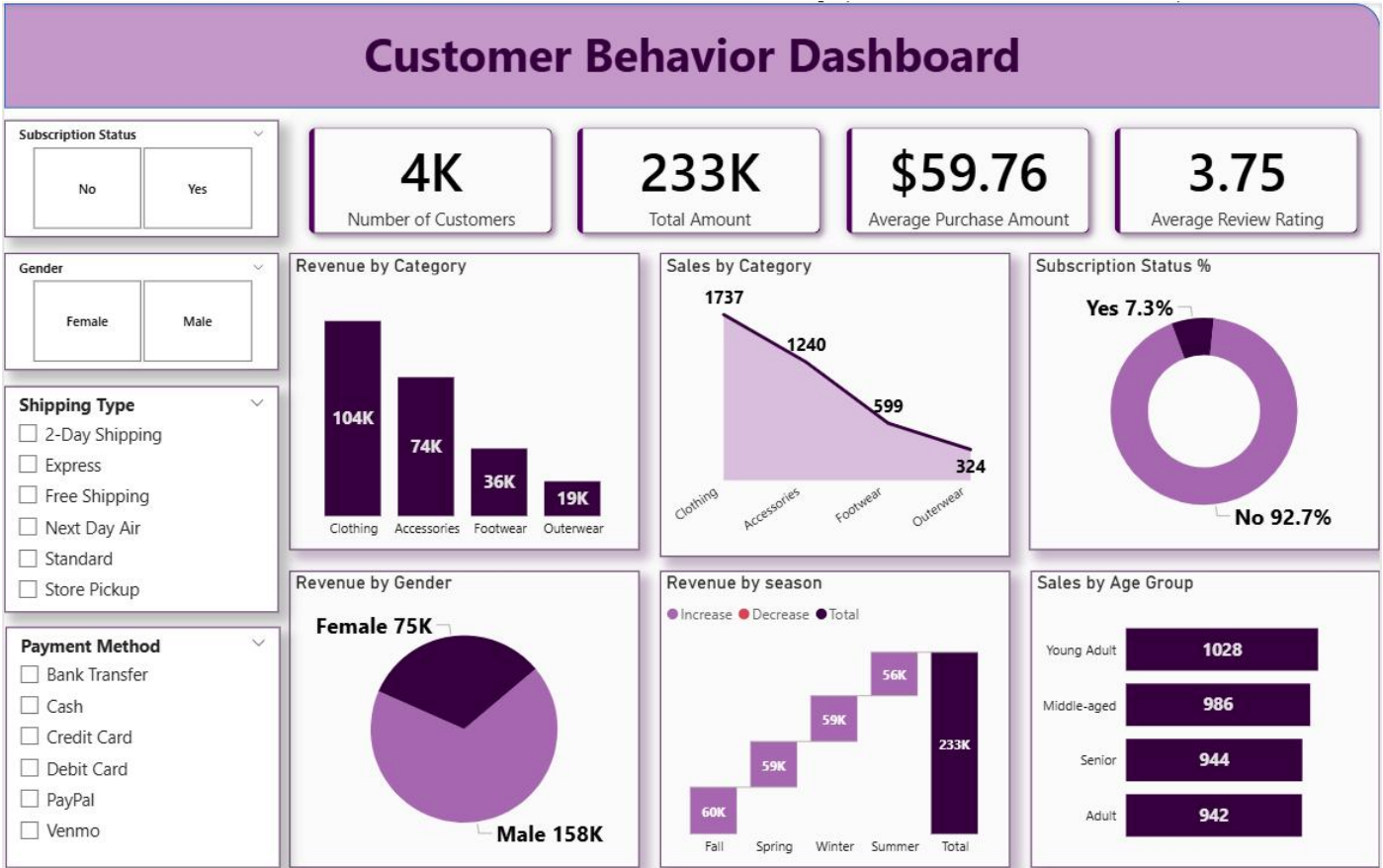
Average Purchase Amount by Age Group : Average purchase amounts were compared across different age groups.

Observation:

- Middle-aged customers tend to spend more per transaction compared to younger customers.

	age_group	revenue
►	Young Adult	62143
	Middle-aged	59197
	Adult	55978
	Senior	55763

5. Dashboard in PowerBI



6. Business Recommendations

- Implement **targeted discount strategies** instead of offering discounts to all customers to protect profit margins.
- Focus on **customer retention strategies** to convert returning customers into loyal customers.
- Promote **high-rated products** more prominently to increase customer trust and sales.
- Strengthen **loyalty and subscription programs** by offering exclusive benefits such as early access or special discounts.
- Encourage higher purchase values by offering **express shipping incentives** for premium orders.
- Use customer insights to improve **inventory planning** and avoid overstocking low-demand products.
- Continuously monitor customer behavior and update strategies using **data-driven analysis**.